# Feature Engineering, Cross-Validation & Model Comparison (Day 3)

UCI Adult (Census Income) · Same hold-out test set as Day 1/2 · Due 2 Sept 2026

## 1. Engineered Feature List

| Feature | Type | Rule | Justification | Mutual Info |
|---|---|---|---|---|
| is_married | bool | marital-status starts 'Married' | Collapses strongest known signal into 1 col | 0.107 |
| edu_hours_interaction | numeric | education-num × hours-per-week | Education pays off more w/ more hours | 0.084 |
| log_capital_gain | numeric | log1p(capital-gain) | Compresses extreme right-skew | 0.081 |
| age_bucket | categorical | 5 age bands | Age-income relationship is nonlinear | 0.060 |
| higher_ed | bool | education-num >= 13 | Turns Day1 rule baseline into a feature | 0.047 |
| hours_bin | categorical | part-time/full-time/OT bands | Captures 40hr/wk step Day1 EDA found | 0.038 |
| has_capital_gain | bool | capital-gain > 0 | Simple flag for zero-inflated column | 0.028 |

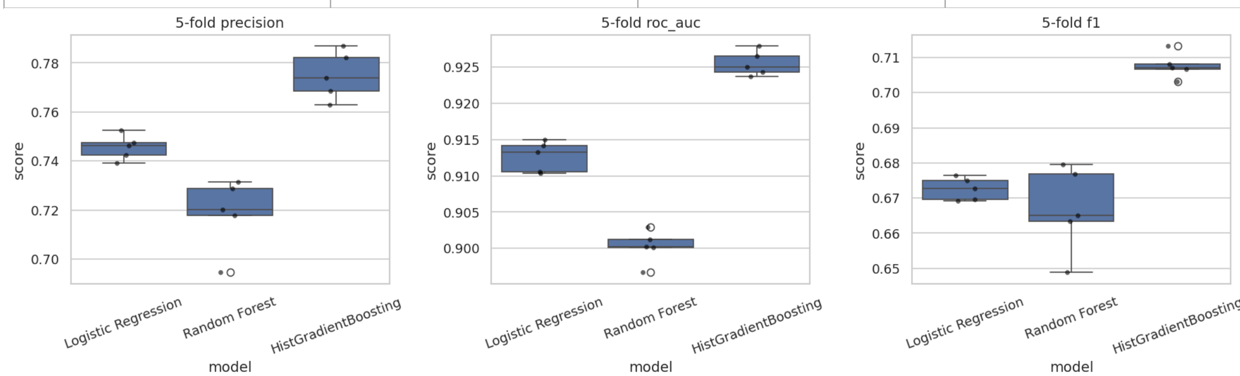
Ranked by mutual information (training set only, no leakage): marital/education/capital-gain-derived features carry the strongest signal, consistent with every prior day's findings. All features are built with a pure row-wise function (no dataset statistics, no target usage) so they apply identically and safely to train/dev/test.

## 2. Pipeline Integration

Feature engineering is wrapped in a **FunctionTransformer** as the first pipeline step, re-run identically inside every CV fold — it only reads values already present in each row, so there is no leakage. Output feeds the same ColumnTransformer style as Day 2 (median+scale for numeric, most_frequent+one-hot for categorical), extended to 119 total columns after encoding.

## 3. Cross-Validated Model Comparison (5-fold StratifiedKFold)

| Model | Precision (mean±std) | ROC AUC (mean±std) | F1 (mean±std) |
|---|---|---|---|
| Logistic Regression | 0.7456 ± 0.0045 | 0.9127 ± 0.0019 | 0.6726 ± 0.0029 |
| Random Forest | 0.7185 ± 0.0130 | 0.9003 ± 0.0021 | 0.6668 ± 0.0110 |
| **HistGradientBoosting** | **0.7749 ± 0.0087** | **0.9255 ± 0.0015** | **0.7077 ± 0.0032** |




**HistGradientBoosting wins on all 3 metrics** with the tightest or near-tightest spread, followed by Logistic Regression; Random Forest trails on precision with the widest fold-to-fold variance (std=0.013).

## 4. Statistical Comparison & Feature Importance

**Paired test, top 2 models** (HistGradientBoosting 0.7749 vs Logistic Regression 0.7456 mean precision): paired t-test t=8.08, **p=0.0013**; Wilcoxon p=0.063. **Interpretation:** the t-test suggests a real difference, but with only 5 folds the Wilcoxon result is more borderline — and the effect size (~0.03 precision gap) is larger than either model's fold std, so we treat this as a **genuine, if modest, edge for HistGradientBoosting** rather than noise, while acknowledging 5 folds gives limited statistical power.
**Random Forest feature importances (top 5):** fnlwgt (0.142), age (0.107), edu_hours_interaction (0.069), marital-status=Married-civ-spouse (0.055), log_capital_gain (0.047). Among engineered features specifically, **edu_hours_interaction, log_capital_gain, and is_married rank highest** — confirming the mutual-information ranking from Task 1. **Logistic Regression coefficients** show is_married (+0.80) and log_capital_gain (+0.50) as top positive engineered features, consistent with the tree model. **Caveat:** has_capital_gain shows a large negative LR coefficient (-3.73) purely due to multicollinearity with log_capital_gain (near-identical information, since one is derived from the other) — a reminder that individual linear coefficients can mislead under correlated features, even when the model's overall predictions are fine.
**Does this make sense?** Yes — engineered features built from marital-status/education/capital-gain inherit the strength those raw columns already showed in Day 1/2, while age_bucket/hours_bin add less because the raw age/hours-per-week columns are still present and largely redundant with their bucketed versions.

## 5. Feature Selection / Dimensionality Check

| k features | Precision (mean) | ROC AUC (mean) | Total fit time (5-fold) |
|---|---|---|---|
| 20 (SelectKBest) | 0.7691 | 0.9203 | 61.7s |
| 40 (SelectKBest) | 0.7745 | 0.9229 | 60.3s |
| All 119 (no selection) | 0.7749 | 0.9255 | 11.7s |

**Recommendation: keep the full 119-feature set for Day 4 tuning.** Selecting fewer features (k=20 or 40) costs a small but real precision drop (-0.006 to -0.058) that exceeds HistGradientBoosting's own fold std (0.0087) at k=20, and — counter-intuitively — selection made total fit time **slower**, not faster (61.7s / 60.3s vs 11.7s), because SelectKBest's mutual-information scoring is itself expensive to recompute every fold. Since precision is our primary business metric and dimensionality reduction didn't even deliver its usual speed benefit here, there's no upside to trade away accuracy for. **Recommended for Day 4:** HistGradientBoosting and Logistic Regression, full engineered feature set, no SelectKBest step.